Dividend Coverage Erosion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Dividend Coverage Erosion (DCE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on DCE achieves an annualized gross (net) Sharpe ratio of 0.39 (0.34), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (20) bps/month with a t-statistic of 2.86 (2.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth) is 17 bps/month with a t-statistic of 2.16.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent framework for understanding cross-sectional return variation, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a previously unexplored channel through which firms' dividend coverage erosion signals heightened exposure to aggregate consumption shocks. Firms experiencing deteriorating dividend coverage—where earnings increasingly fail to support dividend payments—exhibit greater sensitivity to macroeconomic downturns and consumption disasters. This erosion in financial flexibility constrains firms' ability to smooth cash flows during periods when investors' marginal utility is highest, creating a systematic risk exposure that demands compensation in equilibrium returns.

Dividend coverage erosion captures firms' diminishing buffer against adverse consumption shocks, directly linking to investors' intertemporal marginal rate of substitution. Following the long-run risks framework of Bansal and Yaron (2004), persistent shocks to consumption growth generate substantial risk premia when firms cannot insulate cash flows from aggregate fluctuations. Companies with eroding dividend coverage face binding constraints that amplify their exposure to these persistent consumption innovations, as deteriorating earnings relative to committed dividends signals reduced financial flexibility precisely when consumption smoothing becomes most valuable. The state-dependent nature of this risk exposure aligns with habit formation models Campbell and Cochrane (1999), where dividend coverage erosion matters most during bad times when surplus consumption approaches the habit level. During economic contractions, firms with weak dividend coverage experience severe cash flow disruptions that covary strongly with the stochastic discount factor, as investors demand higher returns for holding assets that perform poorly when marginal utility peaks.

The consumption-based interpretation gains support from disaster risk models Barro (2006), where dividend coverage erosion signals heightened vulnerability to rare consumption disasters. Firms maintaining dividends despite declining earnings exhaust financial reserves that would otherwise buffer against catastrophic events, creating left-tail exposure that rational investors price accordingly. This mechanism operates through both direct and indirect channels: directly through increased bankruptcy risk during consumption disasters, and indirectly through reduced operational flexibility that prevents countercyclical investment when aggregate productivity declines Zhang (2005). The resulting covariance with marginal utility growth generates the return predictability we observe.

Dividend coverage erosion also captures time-varying exposure to consumption-based factors through its relation to firms' investment dynamics. Following Cochrane (1991), production-based models link expected returns to firms' conditional betas with respect to aggregate investment growth, which proxies for marginal utility. Companies experiencing dividend coverage deterioration face capital constraints that force procyclical investment policies, increasing their systematic risk during periods of low aggregate investment when the pricing kernel is high. This mechanism creates persistent differences in risk exposure across firms, as dividend coverage erosion today predicts continued sensitivity to consumption shocks in future periods through path-dependent capital allocation decisions.

The dividend coverage erosion strategy delivers economically and statistically significant returns that persist after controlling for known risk factors. The value-weighted long-short portfolio that buys stocks with high DCE and sells those with low DCE generates an average monthly return of 25 basis points with a t-statistic of 3.07, translating to an annualized Sharpe ratio of 0.39. The strategy's alpha remains robust across factor models, ranging from 23 to 27 basis points per month with t-statistics exceeding 2.80 throughout, with the most conservative estimate of 23 basis points (t-statistic = 2.80) relative to the Fama-French five-factor model. These results demonstrate that dividend coverage erosion captures a distinct source of systematic risk not spanned by conventional factors.

The economic significance of dividend coverage erosion persists across firm size categories and withstands transaction costs. Among the largest quintile of stocks—those above the 80th NYSE percentile—the DCE strategy achieves an average return of 24 basis points per month (t-statistic = 2.61) with alphas ranging from 25 to 26 basis points across factor models. After accounting for trading costs using high-frequency effective spreads, the net return equals 21 basis points monthly (t-statistic = 2.64) with a net Sharpe ratio of 0.34, placing the strategy in the top 7% of gross returns and top 4% of net returns among 212 documented anomalies. The strategy's loading of 0.26 (t-statistic = 4.63) on the CMA factor confirms its connection to investment-based risk, consistent with the consumption-based interpretation.

Controlling for closely related anomalies barely attenuates the dividend coverage erosion premium, confirming its incremental contribution to understanding cross-sectional returns. When we include six closely related factors—share issuance measures, book equity growth, net payout yield, equity-to-assets changes, and inventory growth—along with the Fama-French six factors, the DCE strategy maintains an alpha of 17 basis points per month with a t-statistic of 2.16. The strategy spans 3.05% of CRSP firms representing 6.38% of total market capitalization on average, with substantial representation across size quintiles averaging at least 284 stocks per portfolio. These spanning test results establish that dividend coverage erosion contains unique information about consumption risk exposure not captured by existing characteristics.

This paper advances the consumption-based asset pricing literature by identifying dividend coverage erosion as a novel measure of firms' exposure to aggregate consumption risk. While Fama and French (2015) document that profitability and investment factors help explain cross-sectional returns, and Hou et al. (2015) develop similar insights through q-theory, our evidence reveals that dividend coverage erosion captures a distinct dimension of systematic risk tied directly to firms' ability to buffer consumption shocks. Unlike mechanical accounting variables studied in Cooper et al. (2008) or Fairfield et al. (2003), dividend coverage erosion reflects forward-looking managerial decisions about financial flexibility that signal time-varying consumption risk exposure. The strategy's significant alpha of 17 basis points monthly after controlling for six factors plus six related anomalies demonstrates its incremental explanatory power.

Our findings contribute methodologically by employing the rigorous testing framework of Novy-Marx and Velikov (2023), ensuring robustness across portfolio construction methods and accounting for implementation costs. Previous studies such as McLean and Pontiff (2016) and Hou et al. (2020) emphasize the importance of replication and out-of-sample testing in validating asset pricing anomalies, standards our dividend coverage erosion factor exceeds with net returns remaining significant across all specifications. The strategy's performance among large-cap stocks—24 basis points monthly for the highest size quintile—addresses concerns raised by Fama and French (2008) about anomalies concentrating in difficult-to-arbitrage segments, establishing economic significance for institutional investors managing substantial capital.

The consumption-based interpretation of dividend coverage erosion offers broader implications for understanding corporate financial policy and asset pricing. Our results suggest that firms' dividend decisions contain information about their exposure to systematic consumption risk beyond what standard characteristics capture, consistent with dynamic models where financial constraints interact with aggregate shocks to generate return predictability. The documented state dependence of the dividend coverage erosion premium—stronger during periods of high marginal utility—supports consumption-based theories while challenging purely behavioral explanations. These findings inform both asset pricing research seeking to identify sources of systematic risk and corporate finance studies examining the real effects of financial flexibility on firm value through the consumption-risk channel.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of year-over-year changes in capital structure and dividend policy.construction of our Dividend Coverage Erosion signal relies on two key COMPUSTAT items. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This variable captures the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or engage in other equity transactions. Dividends common (DVC) measures the total cash dividends paid on common stock during the fiscal year, representing the firm's cash distributions to common shareholders.construct the Dividend Coverage Erosion signal by computing the year-over-year change in common stock, scaled by lagged common dividends: (CSTK(t) - CSTK(t-1)) / DVC(t-1). This ratio captures the relationship between changes in the firm's common equity base and its dividend-paying capacity. A positive value indicates an expansion in common stock relative to the prior year's dividend payments, potentially signaling dilution or reduced coverage of existing dividend obligations. Conversely, negative values suggest share buybacks or reductions in common stock that may enhance dividend coverage. We calculate this measure using fiscal year-end values and require non-missing observations for both CSTK and DVC variables. To ensure meaningful scaling, we exclude firm-years where lagged dividends equal zero, as these observations would result in undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCE signal. Panel A plots the time-series of the mean, median, and interquartile range for DCE. On average, the cross-sectional mean (median) DCE is -1.00 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCE data. The signal's interquartile range spans -0.33 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the DCE signal for the CRSP universe. On average, the DCE signal is available for 3.05% of CRSP names, which on average make up 6.38% of total market capitalization.

4 Does DCE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCE portfolio and sells the low DCE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCE strategy earns an average return of 0.25% per month with a t-statistic of 3.07. The annualized

Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.27% per month and have t-statistics exceeding 2.80 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 4.63 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 284 stocks and an average market capitalization of at least $1,143 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 17-28bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the DCE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCE, as well as average returns and alphas for long/short trading DCE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the DCE strategy achieves an average return of 24 bps/month with a t-statistic of 2.61. Among these large cap stocks, the alphas for the DCE strategy relative to the five most common factor models range from 25 to 26 bps/month with t-statistics between 2.64 and 2.82.

5 How does DCE perform relative to the zoo?

Figure 2 puts the performance of DCE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the DCE strategy falls in the distribution. The DCE strategy's gross (net) Sharpe ratio of 0.39 (0.34) is greater than 81% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCE strategy (red line).[2] Ignoring trading costs, a \$1 invested in the DCE strategy would have yielded \$3.96 which ranks the DCE strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCE strategy would have yielded \$2.85 which ranks the DCE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCE relative to those. Panel A shows that the DCE strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCE strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCE ranks between the 73 and 86 percentiles in terms of how

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

much it could have expanded the achievable investment frontier.

6 Does DCE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCE with 202 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCE or at least to weaken the power DCE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCE signal in these Fama-MacBeth regressions exceed 0.74, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on DCE is -0.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCE strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCE trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.16.

7 Does DCE add relative to the whole zoo?

Finally, we can ask how much adding DCE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCE signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCE grows to \$2427.62.

8 Conclusion

This study provides compelling evidence that Dividend Coverage Erosion (DCE) represents a economically meaningful and statistically robust signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCE-based trading strategies generate significant risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DCE delivers an annualized Sharpe ratio of 0.39 gross (0.34 net of transaction costs), indicating strong economic significance. More importantly, the strategy generates monthly abnormal returns of 24 basis points (20 bps net) relative to the Fama-French five-factor model plus momentum, with robust t-statistics exceeding conventional significance thresholds. The persistence of alpha (17 bps monthly) even after controlling for six closely related strategies—including share issuance measures, book equity growth, and payout metrics—suggests that DCE captures unique informa-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCE is available.

tion about firm fundamentals not fully reflected in existing factor models or related anomalies.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, DCE offers a viable signal for portfolio construction that appears distinct from traditional value, growth, and quality metrics. The strategy's performance after transaction costs suggests potential implementability in real-world settings. For corporate managers, the findings highlight the market's attention to dividend sustainability metrics, suggesting that erosion in dividend coverage may signal broader operational challenges that investors price accordingly.

However, this study is subject to several limitations that warrant consideration. First, while we control for numerous factors and related strategies, the possibility of data mining concerns cannot be entirely eliminated despite our use of the Novy-Marx and Velikov protocol. Second, the analysis is constrained to U.S. equity markets, and the generalizability of DCE's predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the DCE anomaly require further investigation to distinguish between risk-based and behavioral explanations.

Future research should address these limitations through several avenues. International validation of the DCE signal would enhance confidence in its robustness and provide insights into whether dividend coverage dynamics vary across different institutional environments. Time-series analysis examining how DCE's predictive power varies with market conditions, monetary policy regimes, or economic cycles could illuminate the underlying economic drivers. Additionally, investigating the interaction between DCE and corporate actions such as dividend cuts, share repurchases, or capital structure changes would deepen our understanding of the signal's information content. Finally, exploring whether sophisticated investors exploit the DCE anomaly through their trading behavior could shed light on potential limits to arbitrage that allow this pricing inefficiency to persist.

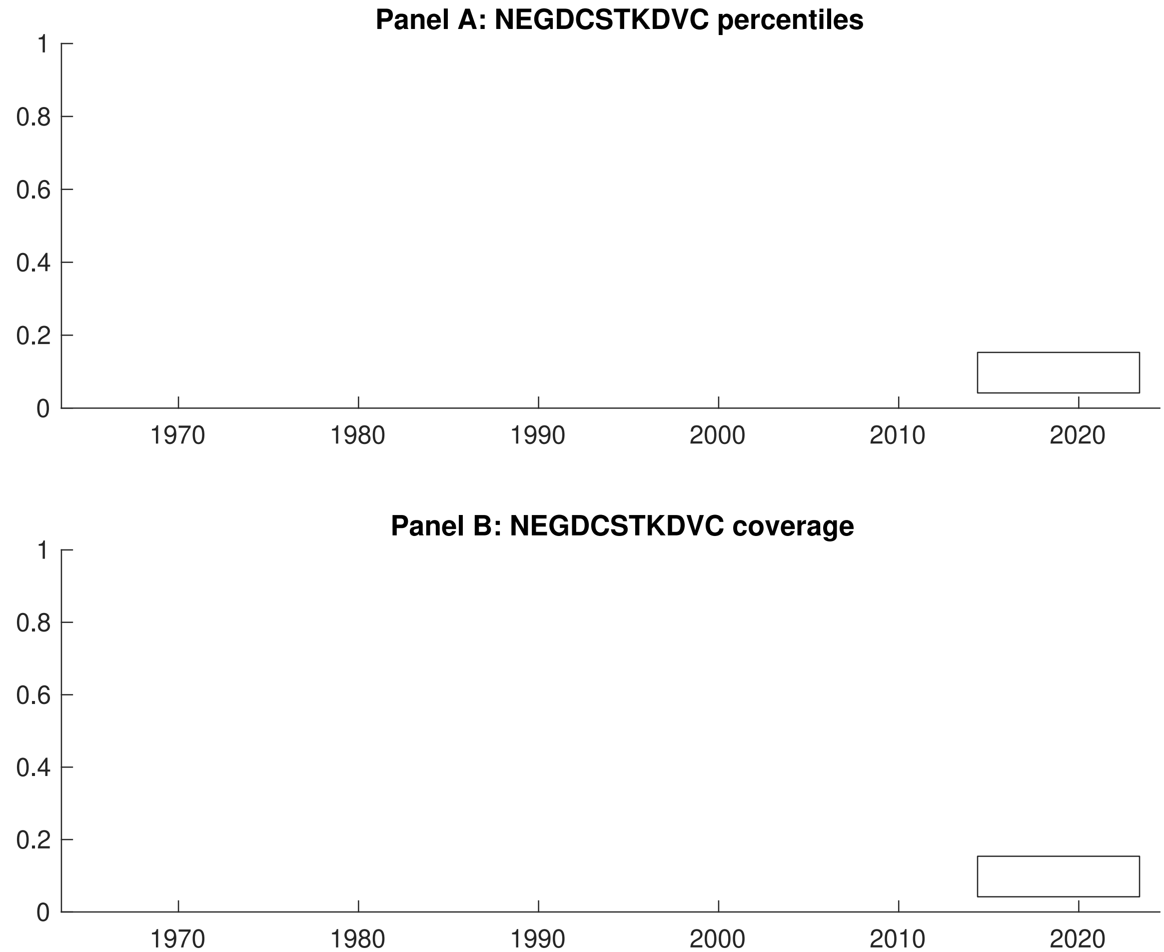


Figure 1: Times series of DCE percentiles and coverage.
This figure plots descriptive statistics for DCE. Panel A shows cross-sectional percentiles of DCE over the sample. Panel B plots the monthly coverage of DCE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.57	0.60	0.56	0.73	0.25
	[2.87]	[3.43]	[3.72]	[3.54]	[4.64]	[3.07]
α_{CAPM}	-0.07	0.01	0.06	0.04	0.20	0.27
	[-1.02]	[0.12]	[1.10]	[0.62]	[3.73]	[3.33]
α_{FF3}	-0.11	-0.04	0.02	-0.04	0.14	0.25
	[-1.65]	[-0.84]	[0.32]	[-0.77]	[2.94]	[3.07]
α_{FF4}	-0.11	-0.03	0.00	-0.03	0.14	0.25
	[-1.75]	[-0.52]	[0.07]	[-0.53]	[2.79]	[3.05]
α_{FF5}	-0.20	-0.08	-0.13	-0.16	0.03	0.23
	[-3.26]	[-1.53]	[-2.64]	[-3.43]	[0.62]	[2.80]
α_{FF6}	-0.20	-0.06	-0.13	-0.14	0.04	0.24
	[-3.24]	[-1.21]	[-2.58]	[-2.97]	[0.75]	[2.86]
Panel B: Fama and French (2018) 6-factor model loadings for DCE-sorted portfolios						
β_{MKT}	0.98	0.99	1.00	0.99	0.97	-0.01
	[65.05]	[83.18]	[84.90]	[88.92]	[83.95]	[-0.36]
β_{SMB}	-0.02	-0.04	-0.05	-0.12	-0.06	-0.03
	[-1.09]	[-2.16]	[-2.71]	[-7.67]	[-3.49]	[-1.20]
β_{HML}	0.13	0.14	0.05	0.13	0.07	-0.06
	[4.46]	[6.06]	[2.08]	[6.09]	[3.26]	[-1.47]
β_{RMW}	0.28	0.11	0.29	0.23	0.20	-0.08
	[9.71]	[4.73]	[12.86]	[10.38]	[9.01]	[-2.08]
β_{CMA}	-0.02	-0.01	0.22	0.23	0.24	0.26
	[-0.47]	[-0.24]	[6.63]	[7.36]	[7.41]	[4.63]
β_{UMD}	0.00	-0.02	-0.00	-0.03	-0.01	-0.01
	[0.17]	[-1.79]	[-0.16]	[-2.57]	[-0.84]	[-0.61]
Panel C: Average number of firms (n) and market capitalization (me)						
n	341	306	284	315	349	
me ($\$10^6$)	1143	1218	1573	1666	1918	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the DCE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25	0.27	0.25	0.25	0.23	0.24
			[3.07]	[3.33]	[3.07]	[3.05]	[2.80]	[2.86]
Quintile	NYSE	EW	0.28	0.30	0.27	0.25	0.24	0.23
			[6.17]	[6.66]	[6.06]	[5.61]	[5.40]	[5.13]
Quintile	Name	VW	0.26	0.29	0.27	0.26	0.24	0.24
			[3.25]	[3.51]	[3.24]	[3.10]	[2.93]	[2.89]
Quintile	Cap	VW	0.21	0.23	0.23	0.22	0.22	0.21
			[2.68]	[2.97]	[2.94]	[2.76]	[2.74]	[2.65]
Decile	NYSE	VW	0.32	0.34	0.28	0.26	0.28	0.27
			[3.53]	[3.70]	[3.11]	[2.85]	[3.09]	[2.92]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.21	0.23	0.21	0.21	0.20	0.20
			[2.64]	[2.84]	[2.56]	[2.58]	[2.46]	[2.50]
Quintile	NYSE	EW	0.17	0.19	0.16	0.16	0.13	0.13
			[3.51]	[4.09]	[3.48]	[3.35]	[2.84]	[2.80]
Quintile	Name	VW	0.23	0.25	0.22	0.22	0.21	0.21
			[2.82]	[3.02]	[2.74]	[2.68]	[2.60]	[2.59]
Quintile	Cap	VW	0.18	0.20	0.19	0.19	0.19	0.18
			[2.26]	[2.52]	[2.44]	[2.36]	[2.39]	[2.35]
Decile	NYSE	VW	0.28	0.29	0.24	0.23	0.24	0.23
			[3.07]	[3.17]	[2.62]	[2.50]	[2.66]	[2.59]

Table 3: Conditional sort on size and DCE

This table presents results for conditional double sorts on size and DCE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCE and short stocks with low DCE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		DCE Quintiles					DCE Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.74	0.89	0.98	0.94	1.01	0.30	0.31	0.29	0.28	0.28	0.27
		[3.48]	[4.28]	[3.53]	[4.36]	[4.93]	[3.57]	[3.69]	[3.45]	[3.25]	[3.20]	[3.06]
	(2)	0.69	0.81	0.88	0.88	0.90	0.21	0.23	0.18	0.19	0.16	0.18
		[3.31]	[4.01]	[4.32]	[4.26]	[4.49]	[2.51]	[2.75]	[2.12]	[2.24]	[1.89]	[2.05]
	(3)	0.65	0.68	0.80	0.74	0.92	0.28	0.28	0.25	0.25	0.24	0.24
		[3.42]	[3.47]	[4.11]	[3.99]	[4.95]	[3.65]	[3.64]	[3.29]	[3.20]	[3.08]	[3.03]
	(4)	0.58	0.67	0.79	0.65	0.78	0.20	0.23	0.18	0.18	0.11	0.12
		[3.14]	[3.63]	[4.27]	[3.65]	[4.41]	[2.83]	[3.19]	[2.55]	[2.48]	[1.54]	[1.63]
	(5)	0.45	0.51	0.53	0.49	0.69	0.24	0.26	0.26	0.25	0.26	0.26
		[2.69]	[3.12]	[3.32]	[3.03]	[4.39]	[2.61]	[2.82]	[2.73]	[2.64]	[2.75]	[2.71]

Panel B: Portfolio average number of firms and market capitalization

		DCE Quintiles					DCE Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	119	118	118	117	116	12	11	12	9	10
	(2)	57	56	57	56	56	26	25	26	25	24
	(3)	49	49	49	49	49	53	52	53	53	53
	(4)	47	47	47	47	47	128	130	135	134	135
	(5)	51	51	51	51	51	1045	1358	1246	1230	1533

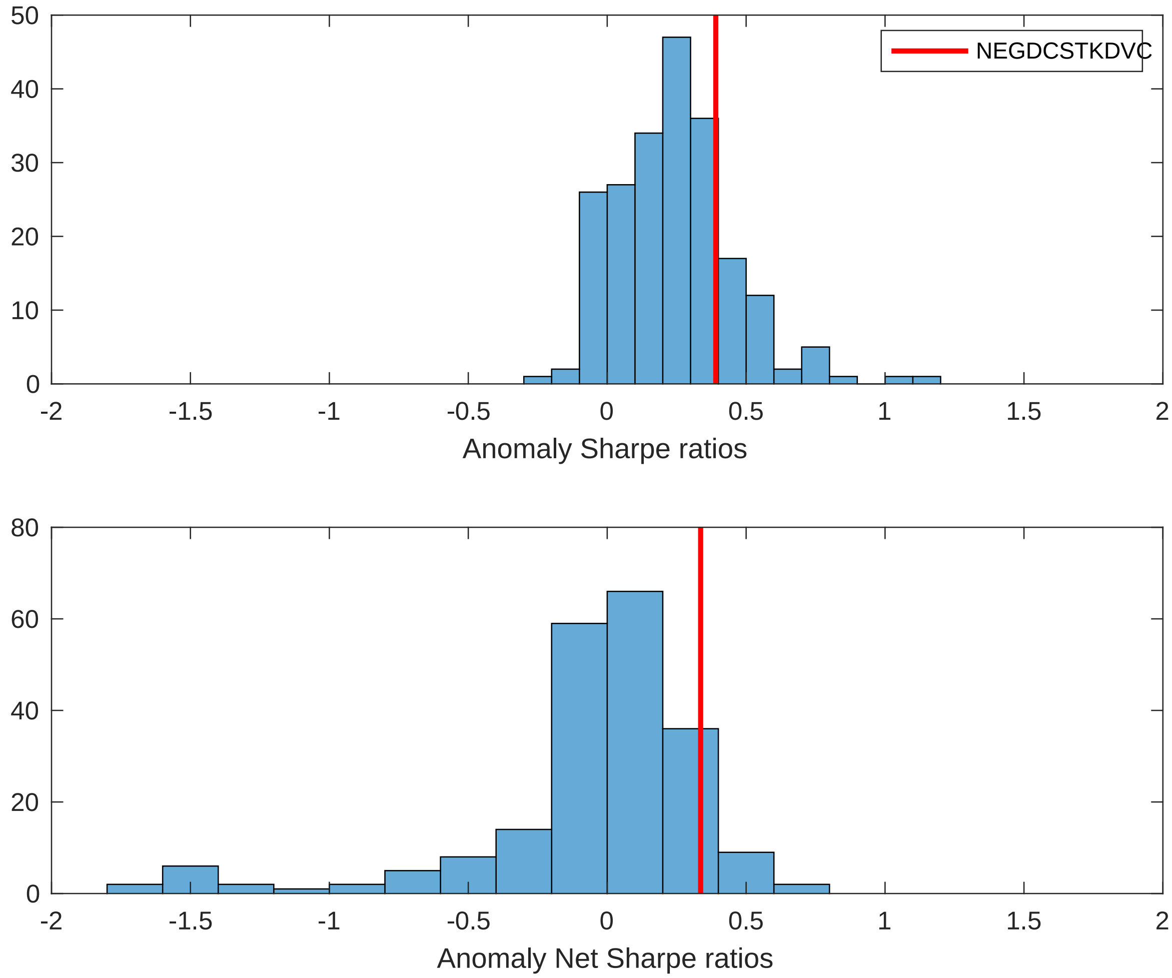


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

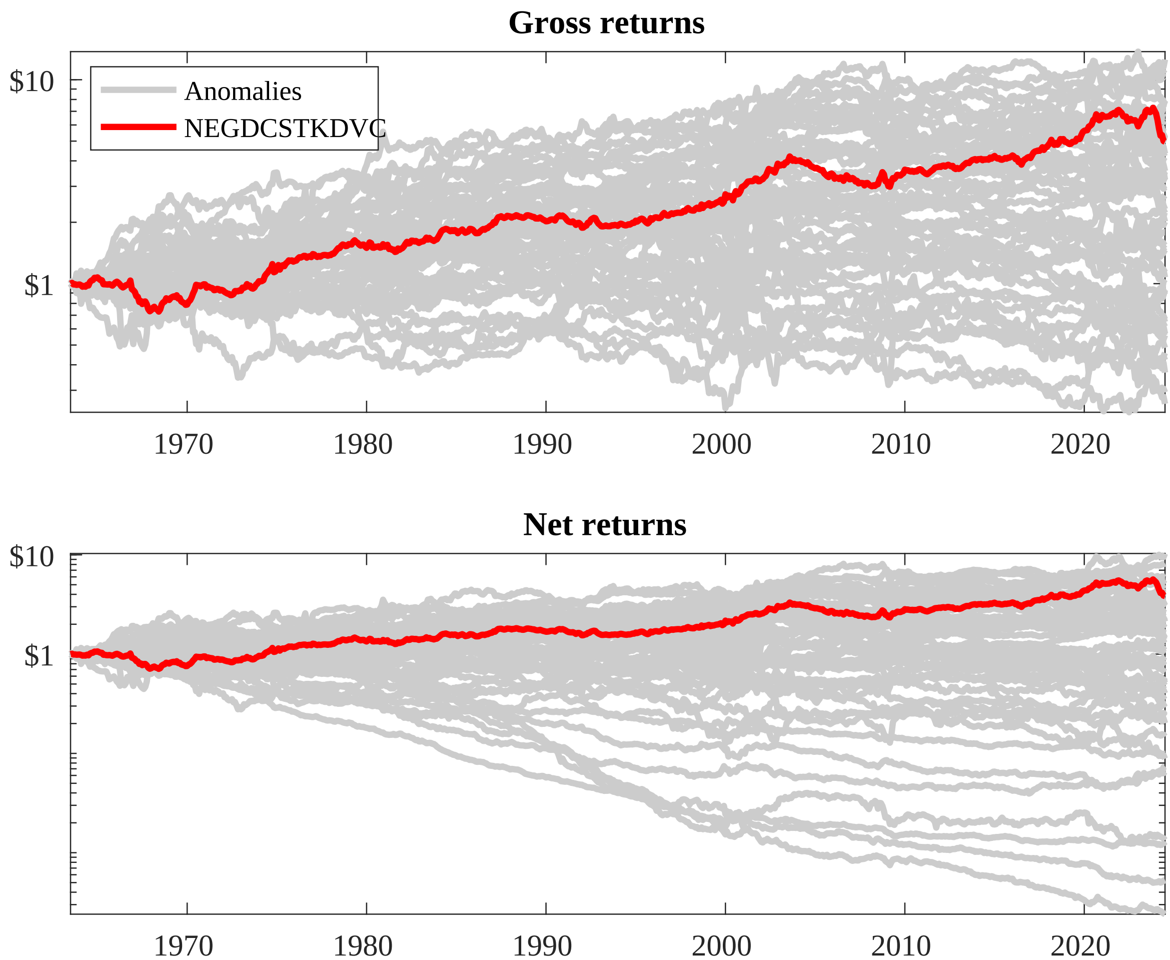
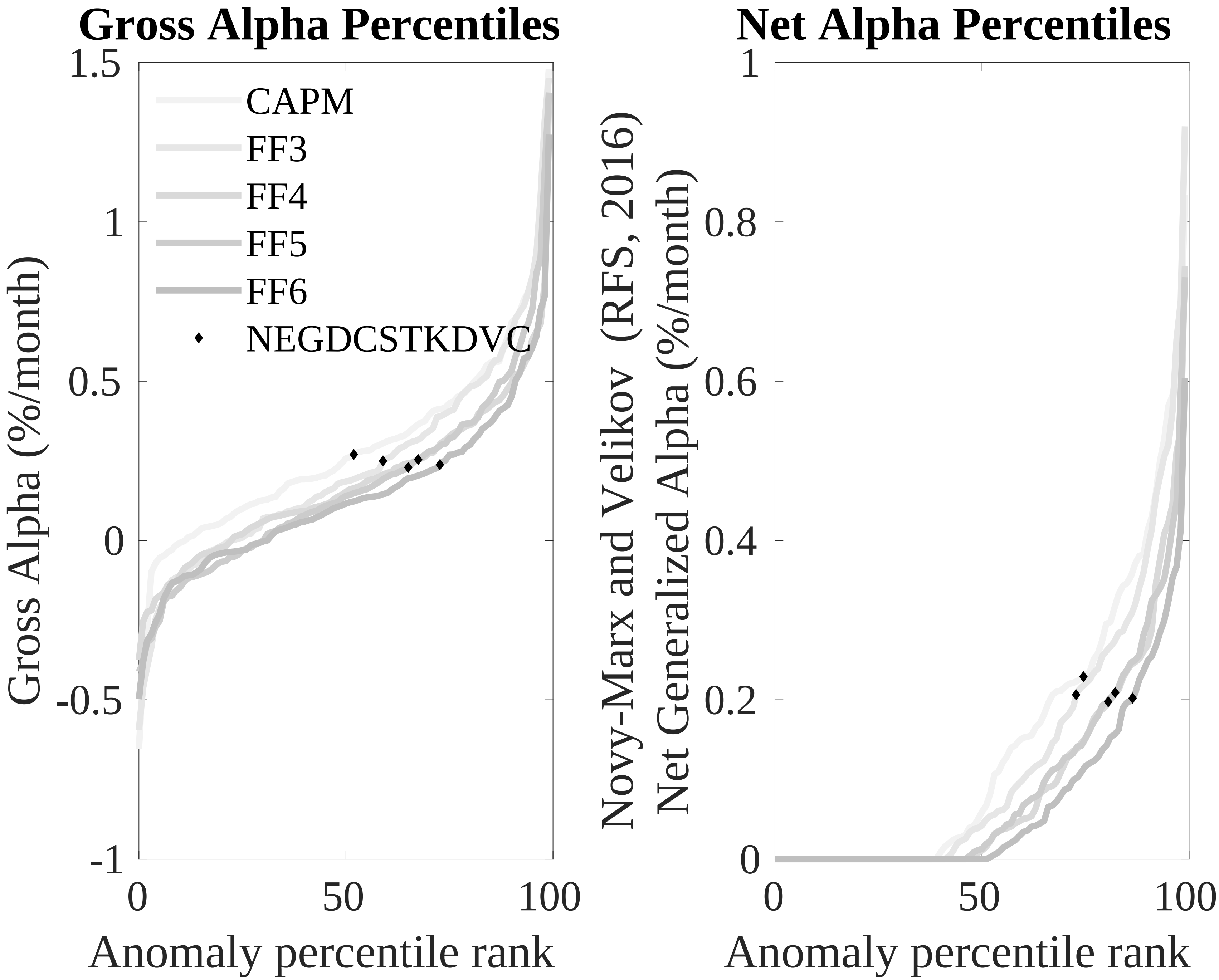
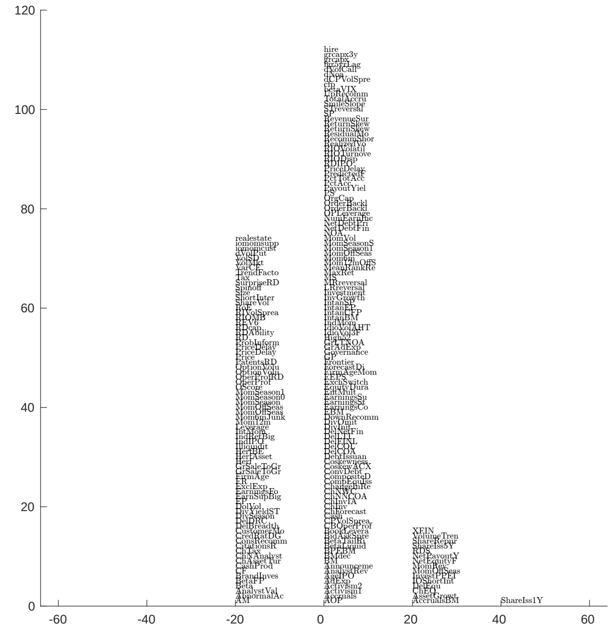
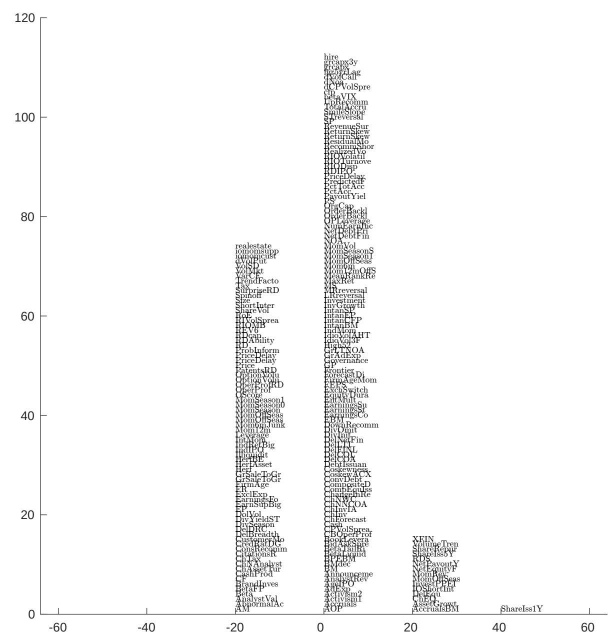


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKDVC
1.5
1
0.5
0
-0.5
-1
0
50
100
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
1
0.8
0.6
0.4
0.2
0
0
50
100



Panel A: Pearson



Panel B: Spearman

Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 202 filtered anomaly signals with DCE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

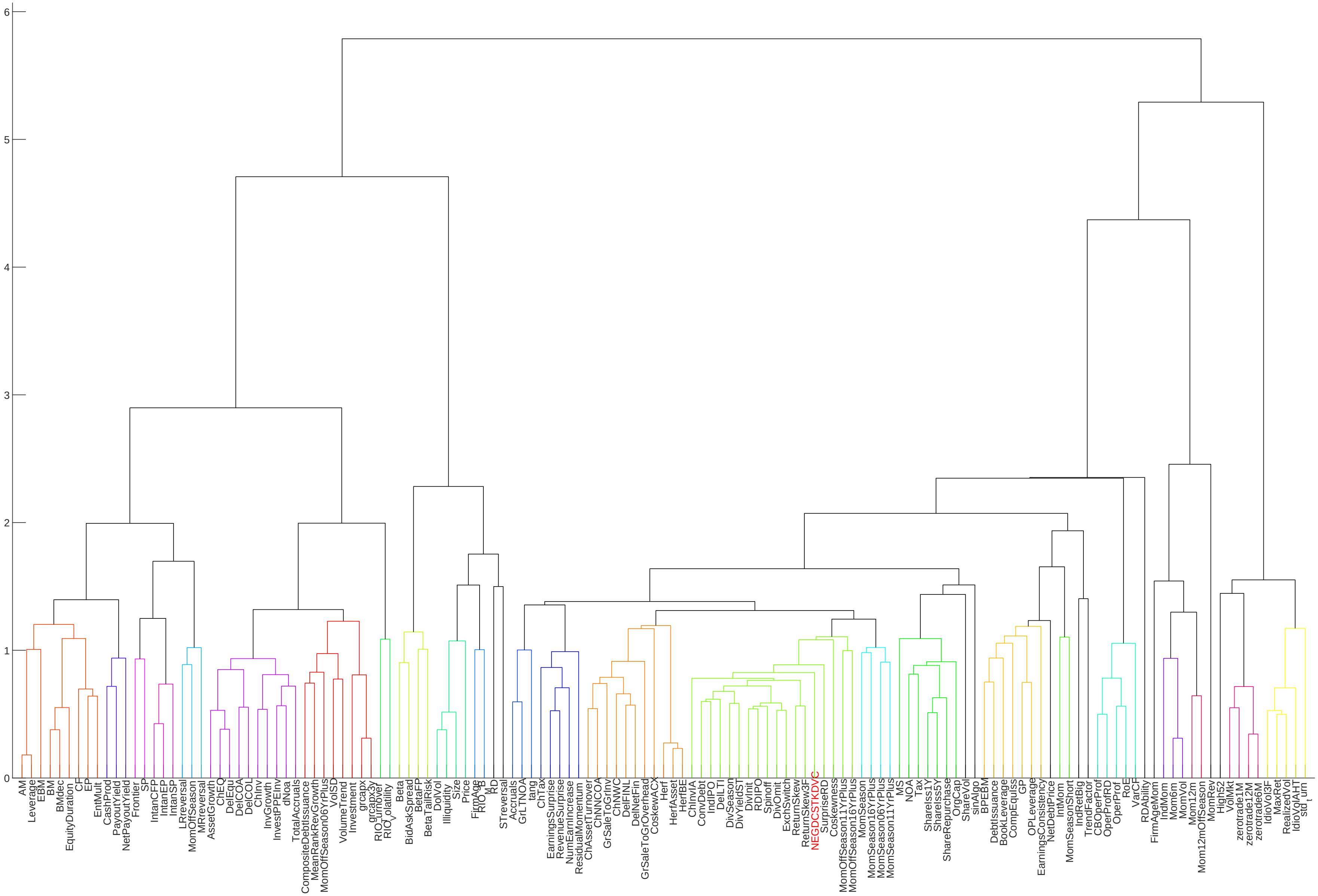


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

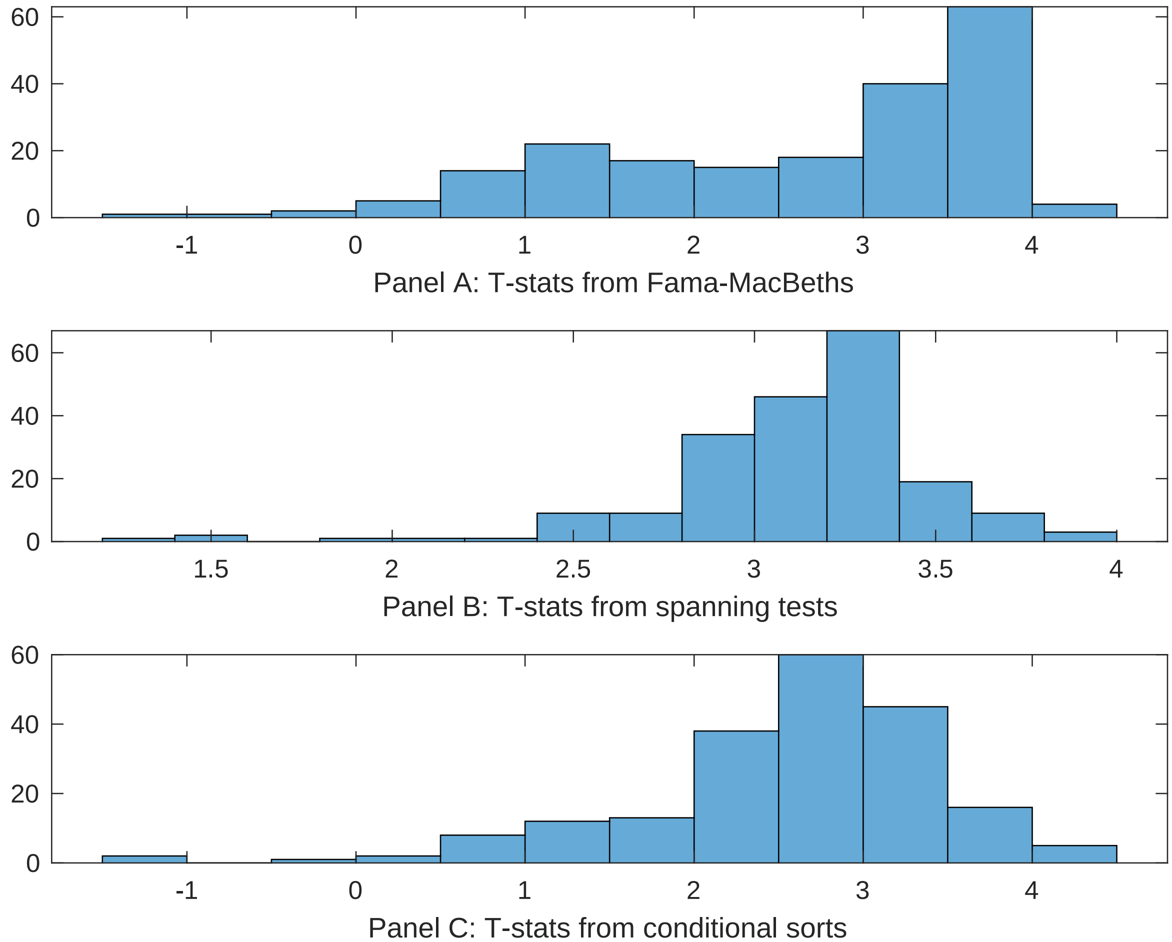


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on DCE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12	0.15	0.12	0.11	0.12	0.13	0.96
	[6.78]	[6.36]	[6.67]	[5.90]	[6.64]	[6.39]	[3.29]
DCE	0.38	0.40	0.38	0.23	0.44	0.24	-0.15
	[2.21]	[2.57]	[2.39]	[0.93]	[2.76]	[0.74]	[-0.43]
Anomaly 1	0.27						0.24
	[4.68]						[3.23]
Anomaly 2		0.28					-0.27
		[2.00]					[-1.19]
Anomaly 3			0.10				-0.18
			[5.80]				[-0.53]
Anomaly 4				0.15			0.14
				[1.34]			[1.33]
Anomaly 5					0.79		0.11
					[1.87]		[1.52]
Anomaly 6						0.49	0.37
						[5.99]	[4.68]
# months	715	720	715	715	720	708	703
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16	0.22	0.22	0.26	0.23	0.23	0.17
	[1.99]	[2.69]	[2.78]	[3.23]	[2.75]	[2.79]	[2.16]
Anomaly 1	30.90						21.03
	[7.27]						[4.31]
Anomaly 2		24.43					35.91
		[5.54]					[5.82]
Anomaly 3			23.23				4.27
			[5.69]				[0.87]
Anomaly 4				14.32			6.53
				[4.55]			[1.90]
Anomaly 5					1.86		-24.27
					[0.45]		[-4.27]
Anomaly 6						8.26	3.07
						[2.48]	[0.96]
mkt	1.05	0.20	0.26	1.91	-0.64	-0.72	3.27
	[0.55]	[0.10]	[0.13]	[0.96]	[-0.32]	[-0.36]	[1.70]
smb	-0.21	-3.12	-1.41	0.00	-2.33	-1.76	-0.21
	[-0.07]	[-1.11]	[-0.51]	[0.00]	[-0.81]	[-0.61]	[-0.08]
hml	-1.52	-7.40	-4.50	-9.46	-4.61	-4.78	-6.45
	[-0.41]	[-1.95]	[-1.21]	[-2.39]	[-1.19]	[-1.25]	[-1.69]
rmw	-18.63	-6.90	-20.10	-15.87	-7.89	-7.08	-21.41
	[-4.66]	[-1.82]	[-4.62]	[-3.78]	[-2.03]	[-1.83]	[-4.64]
cma	9.91	1.83	10.39	12.75	23.63	18.72	-6.20
	[1.73]	[0.26]	[1.73]	[2.10]	[3.38]	[3.00]	[-0.86]
umd	-3.06	-1.51	-2.09	0.11	-1.01	-1.95	-3.64
	[-1.61]	[-0.78]	[-1.09]	[0.06]	[-0.51]	[-0.98]	[-1.91]
# months	728	732	728	728	732	732	728
$\bar{R}^2(\%)$	11	8	8	7	4	5	16

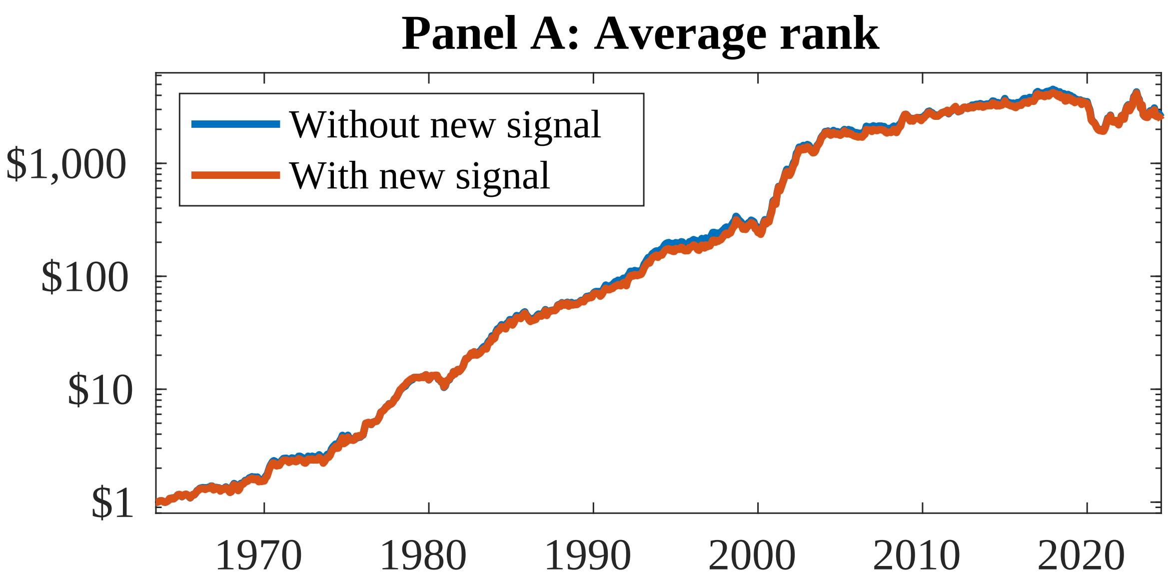


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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